

DAGUPAN, Philippines

WMO: 983250

Lat: 16.0868N

Lon: 120.3520E

Elev: 7

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	70.0	71.4	63.8	88.5	73.3	65.7	94.9	73.6	22.6	78.3	18.7	80.8	4.1	180	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.1	94.7	82.9	93.2	82.8	91.9	82.5	85.5	91.0	84.6	90.2	83.9	89.5	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.2	179.5	89.8	83.3	174.0	89.0	82.4	169.0	88.3	50.1	91.2	49.0	90.1	48.1	89.7	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	15.7	13.3	65.3	99.4	3.6	2.1	62.7	100.9	60.5	102.1	58.5	103.3	55.9	104.8		
(5)			62.3	86.8	3.4	1.9	59.9	88.2	57.9	89.3	56.0	90.4	53.5	91.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	79.1	80.2	82.3	84.9	85.1	83.9	82.4	81.9	82.4	82.6	81.9	80.3		
	DBStd	2.76	2.10	2.15	2.09	2.02	2.35	2.44	2.44	2.42	1.92	1.76	1.77	2.07		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11773	901	845	1001	1048	1088	1018	1004	987	972	1011	957	941		
	CDD65	6298	436	425	536	598	623	568	539	522	522	546	507	476		
	CDH74	68706	3851	4075	5848	7403	7835	6621	5811	5411	5590	6077	5489	4695		
(13)	CDH80	25405	1076	1313	2239	3381	3643	2707	1985	1643	1853	2175	1916	1474		
(14)	Wind	WSAvg	5.6	5.5	5.9	6.4	6.7	6.1	5.4	5.1	5.5	6.2	4.8	4.8		
(15) Precipitation	PrecAvg	99.7	0.3	0.3	0.8	2.4	9.1	14.1	20.9	22.6	13.7	7.8	2.3	0.7		
	PrecMax	183.5	3.1	3.5	5.2	8.7	37.0	44.2	104.7	59.1	37.2	31.8	14.1	5.4		
	PrecMin	49.4	0.0	0.0	0.0	0.0	2.1	2.5	0.5	0.7	1.2	0.3	0.0	0.0		
	PrecStd	28.7	0.6	0.6	1.1	1.9	6.9	9.0	16.3	14.0	7.0	7.0	2.8	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	92.2	94.5	97.0	97.1	95.1	92.6	91.5	91.7	91.7	91.5	90.6		
		MCWB	79.8	80.5	80.9	82.3	83.0	84.5	83.1	83.1	83.4	82.5	81.6	81.2		
	2%	DB	88.0	89.9	92.3	94.6	94.4	93.1	90.8	89.6	89.8	90.2	89.9	88.8		
		MCWB	79.2	80.1	81.0	82.5	83.3	83.9	82.5	82.6	82.4	82.0	81.3	80.6		
	5%	DB	86.5	88.2	90.4	92.8	92.8	91.5	89.4	88.3	88.6	89.1	88.7	87.4		
		MCWB	78.7	79.8	80.7	82.3	83.1	83.3	82.2	82.0	82.0	81.7	81.0	79.9		
	10%	DB	85.1	86.5	88.6	91.1	91.3	90.0	88.1	87.0	87.4	87.9	87.5	86.1		
		MCWB	78.1	79.0	80.2	81.9	82.8	82.7	81.6	81.3	81.6	81.3	80.7	79.4		
	0.4%	WB	82.8	84.3	85.2	86.3	86.7	86.8	85.1	85.0	85.2	84.9	84.8	83.7		
		MCDB	87.3	88.3	90.6	92.6	93.1	92.4	90.3	89.3	89.4	89.3	88.3	88.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	81.0	82.4	83.4	84.6	85.2	85.2	83.6	83.4	83.5	83.5	83.1	82.2		
		MCDB	86.1	87.7	89.2	90.9	91.7	91.1	88.9	88.2	88.2	88.3	87.7	86.9		
	5%	WB	79.8	80.9	81.9	83.5	84.0	84.1	82.6	82.4	82.6	82.5	81.9	80.9		
		MCDB	85.1	86.4	88.1	90.1	90.8	90.1	88.3	87.5	87.5	87.8	87.2	86.1		
	10%	WB	78.8	79.7	80.8	82.5	83.0	82.9	81.7	81.5	81.7	81.7	81.0	79.9		
		MCDB	84.2	85.3	87.1	89.5	89.7	88.9	87.1	86.3	86.5	86.9	86.5	85.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	13.7	14.1	14.2	13.1	11.8	10.2	8.9	10.1	11.1	11.8	12.3		
		MCDBR	15.0	16.3	16.8	16.6	15.6	14.3	12.8	11.6	12.1	13.0	13.7	14.0		
		MCWBR	9.4	9.8	9.1	7.9	7.6	7.3	6.7	6.4	6.7	7.1	7.6	8.6		
	5% WB	MCDBR	13.9	14.7	15.0	14.7	14.3	13.6	12.2	11.0	11.5	12.2	12.7	12.9		
		MCWBR	10.1	10.6	9.8	8.7	8.1	7.9	6.9	6.6	6.9	7.5	8.2	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.405	0.419	0.468	0.520	0.488	0.470	0.470	0.476	0.472	0.481	0.425	0.415		
	taud		2.430	2.379	2.254	2.120	2.246	2.312	2.313	2.286	2.287	2.243	2.419	2.428		
	Ebn at Noon		280	282	270	255	260	262	262	263	265	260	272	274		
	Edh at Noon		35	39	45	51	45	42	42	43	43	44	36	35		
(44) All-Sky Solar Radiation	RadAvg		1510	1718	1903	2023	1827	1670	1490	1398	1512	1552	1453	1377		
	RadStd		64	104	84	89	144	191	195	153	133	121	124	85		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.68	N/A	N/A	+0.72	+1.28	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.58	+1.03	N/A	+0.72	+0.65	+0.76	N/A	N/A	+284

Nomenclature: See separate page